



# Cybersecurity Perspective In Quantum Computing

Along with the immense promise of quantum computing come the enhanced security threats too. This has opened up a whole new field known as post quantum security



Pradip Mukhopadhyay

**“N**ot only is the Universe stranger than we think, it is stranger than we can think,” opined Werner Heisenberg in his famous book ‘Across the Frontiers.’ This quote aptly applies to a quantum computer, which has amazed computer enthusiasts about its promise and ability to solve the otherwise non-solvable, hard problems in our classical computing world.

Quantum computer promises to speed up every individual instruction processing, thus defying Moore’s Law. But it also promises to quickly solve the exponentially repetitive tasks, such as finding out two prime factors

of a given big integer number, which makes it an impending threat for our known cybersecurity universe—a threat to computer’s hardware, software, and stored information. It is a potential threat to our existing cybersecurity concepts and techniques. In this article we shall review this threat to gain a deeper comprehension, so that we can build an opinion around that independently.

Fig. 1 and Fig. 2 indicate the potential of the so-called quantum threat and the global investment behind it; seemingly China is leading the pack.

## Existing cybersecurity techniques

One obvious method of cracking an encryption code is by brute force; keep

trying all possible combinations till the code is broken. But this brute force method is not very practical as it may take years to try out all the combinations of a key. So, the current state of affairs depends on public key encryption. Some well-known techniques are: Diffie-Hellman, elliptic curve, and RSA.

Let us look at RSA. It is related to getting the private key by continuously factoring the public key, a big integer, into product of two primes. Till now this is non-trackable in classical computer paradigm. Because it starts with a big enough integer, it takes years to get the private key out, when compromised. Roughly, the key generation algorithm can be stated as below:

1. Get two equal-size random prime numbers, say  $p$  and  $q$ .